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Evaluation of single vs. staged mesophilic anaerobic digestion of kitchen waste with and without microwave pretreatment

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ABSTRACT

Effects of single and dual stage (acidogenic–methanogenic) mesophilic anaerobic digestion (AD) of kitchen waste (KW) was evaluated at hydraulic retention times (HRTs) of 20, 15, 12 and 9 d with and without thermal microwave (MW) pretreatment (145 °C). Anaerobic acidification in terms of acid accumulation was superior compared to microaerophilic acidification. Maximum anaerobic acidification of KW was determined to occur with an HRT of 2 d which was then selected for the acidification stage. The dual stage AD system fed with untreated KW produced the maximum biogas and volatile solids (VS) stabilization efficiencies at the shortest HRT of 9 d. Conversely, for free liquid resulting from MW pretreatment of KW the two stage reactor at 20 d HRT produced three fold more methane compared with the untreated free liquid control. However, MW pretreatment and AD of the free liquid fraction only, was not a sustainable treatment option. For KW, staging of the AD process had a greater positive impact on waste stabilization and methane yield compared to single stage reactors or MW pretreatment. KW can be characterized as being a readily biodegradable solid waste; concomitantly it is recommended that digester staging without MW pretreatment be employed to maximize methane yield and production.

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1. Introduction

Solid waste management (SWM) has become a major issue in the last decade due to both environmental and economical concerns. Landfilling is still the main approach for solid waste disposal, and about 50% of the waste generated goes directly to landfill in the US. In 2009, only 2.5% of food scraps was recovered while the remaining was landfilled (USEPA, 2009). Kitchen waste (KW) is a highly biodegradable moisture rich residual that is a potential source of environmental contamination, due to the fugitive greenhouse gas emissions and leachate generation within a landfill environment.

Biological processes such as anaerobic digestion (AD) offer sustainable methods to address the problems that may be caused by disposal of organic fraction municipal solid waste (OFMSW) in landfills and has great potential for controlled biogas and energy production. Life Cycle Assessment (LCA) of different waste disposal strategies by Cherubini et al. (2009) including landfilling, AD and waste incineration showed that landfilling is the least favourable option while AD was likely the best option for SWM. Comparing different studies related to composting and AD of OFMSW, Mata-Alvarez (2003) concluded that AD will gain more attention in the future for ecological reasons in particularly less fugitive green house gas emissions and stabilized organic matter residue. Moreover, AD does not consume oxygen, has lower nutrient requirements and it generates methane gas that can be used as a source of energy. The rate limiting step in AD of solid wastes is the hydrolysis of particulate substrates (Eastman and Ferguson, 1981). To accelerate hydrolysis and consequently the entire AD process, solid waste could be pretreated prior to AD, to potentially increase the biogas yield as well as reduce the reaction time and volume of residual solids for final disposal.

Several pretreatment techniques studied in the past (e.g. thermal, chemical, and mechanical) have been shown to effectively enhance AD. Microwave (MW) pretreatment is a novel technology that looks promising. Compared to conventional heating it is attractive due to its environmental and energy conservation properties (Decareau, 1985; Kingston and Jassie, 1988), since it eliminates heat losses that occur in energy transmission during normal heating. In microwave heating direct vibration and rotation of dielectric molecules is possible so localized super heating maximizes heat transfer. Also the existence of the MW athermal effect on enhanced digestion of waste activated sludge (WAS) has been demonstrated by Coelho et al. (2011) and Eskicioglu et al. (2007b).

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Nomenclature

AD	Anaerobic Digestion
BMP	Biochemical Methane Potential
CBP	Cumulative Biogas Production
COD	Chemical Oxygen Demand
F/M	Food to Microorganism ratio
HRT	Hydraulic Retention Time
KW	Kitchen Waste
LCA	Life Cycle Assessment
MAD	Mesophilic Anaerobic Digestion
MW	Microwave
OFMSW	Organic Fraction of Municipal Solid Waste
OLR	Organic Loading Rate
RCF	Relative Centrifugal Force
SCOD	Soluble COD
SKW	Synthetic Kitchen Waste
SRT	Solids Retention Time
SS	Suspended Solids
SWA	Supplement Water Addition
SWA20	SWA of 20%
TCOD	Total COD
TS	Total Solids
TWAS	Thickened Waste Activated Sludge
VFA	Volatile Fatty Acid
VS	Volatile Solids
VVM	Volume Per Volume Per Minute
WAS	Waste Activated Sludge

So far, publications regarding the effect of MW pretreatment on waste solubilisation for enhancing AD have focused on WAS (Hong et al., 2006; Eskicioglu et al., 2008).

Eskicioglu et al. (2007a) performed comparative studies using thickened WAS (TWAS). TWAS characteristics and subsequent enhanced mesophilic anaerobic digestion (MAD) were compared for MW and conventional heating (CH) at two final temperatures (50 and 96 °C) and 3 HRTs (5, 10 and 20 d). The SCOD/TCOD ratio of TWAS pretreated to 96 °C increased from 7% (control) to 15% and 20% for MW and CH pretreatments. Maximum improvements in biogas production and VS removal were observed for 96 °C when the HRT was reduced to 5 d. MW and CH achieved 23 and 26% higher VS removal respectively compared to the control. Similar results for biogas production were achieved, 30 and 35% increases in biogas production over the control for MW and CH.

It has been reported by Mataalvarez et al. (1992) that problems will arise in one-step reactors digesting OFMSW that is readily degradable like KW. The accumulation of VFA in a single reactor can cause imbalances between the methanogenic and acidogenic populations that can result in reduced methane production, thereby reducing the overall efficiency of the reactor. A two-stage acidogenic/methanogenic system may be a solution to overcome the microbial consortia imbalance discussed by different researchers.

Coelho et al. (2011) investigated the effects of MW pretreatment to 96 °C on AD of TWAS in one and two stages reactor under both thermophilic and mesophilic conditions at four HRTs (5, 10, 15, 20 d). The SCOD/TCOD ratio of the TWAS increased from 6% to 20% after MW pretreatment. In two steps AD reactors, with both stages operating under thermophilic condition and overall sludge retention time (SRT) of 5 d, a 106% enhancement biogas was observed compared to the one stage mesophilic control reactor (no MW pretreatment). The single stage mesophilic and thermophilic reactors that were fed with pretreated sludge had 44 and 83% higher biogas production compared to control at HRT of 20 d. In this study the MW pretreatment efficiency was reported to decrease with shortening HRTs for one stage reactors, but efficiency was maintained or increased for two stage reactors.

Observations of Raynal et al. (1998), Schober et al. (1999), Pavan et al. (2000), Bouallagui et al. (2004), Wang et al. (2005) and Liu et al. (2008) confirm that the two-stage anaerobic process for complex wastes is potentially superior to a one-stage process. Most of these studies used strictly anaerobic conditions and WAS without pretreatment. However, biological hydrolysis of solids can be conducted under aerobic as well as anaerobic conditions. Botheju et al. (2009) observed that hydrolysis rates were higher under aerobic conditions and attributed this to higher enzymes production rates using oxygen as the final electron acceptor. Similarly Johansen and Bakke (2006) also reported that addition of very small amounts of air (microaeration) in to an acidogenic reactor can enhance the hydrolysis rate.

MW pretreatment studies of KW in continuous reactors have not yet been addressed. It is hypothesized that combining MW pretreatment with 2 stages AD will enhance the rate and extent of synthetic KW (SKW) stabilization. Semi-continuous digestion tests will be carried out to obtain a better understanding of the effect of MW pretreatment of KW on the performance of single and dual stage AD reactors. Additionally the study will evaluate the effect of MW pretreatment on the solubilisation of KW and the digestion of the free liquid portion only as an alternative to digestion of the whole KW mixture. The secondary objective is to further increase the rate of hydrolysis and acidification by bioaugmentation with WAS and microaerophilic conditions in the first reactor in dual stage digesters.

2. Methods

2.1. Synthetic KW and MW

SKW simulating kitchen waste was used. Components of SKW were cooked rice (18 wt %), cooked pasta (18 wt %), cabbage (11 wt %), carrot (11 wt %), apple (11 wt %), banana (11 wt %), cooked ground beef (10 wt %), wet dog food (10 wt %). Supplemental water addition (supplement water addition) of 20% (SWA20) was used (Shahriari et al., 2012).

A MW Accelerated Reaction System (Mars 5®, CEM Corporation, 0–1200 W and 2450 MHz frequency) was used to irradiate the SKW. The Mars 5 controls and monitors the power, temperature, and pressure within each reaction vessels up to 250 °C and 34.5 bars. MW intensity was controlled by adjusting the temperature ramp time to achieve the set temperature (145 °C with a ramp of 2.7 °C/min).

2.2. Acidogenic fermentation

One of the goals of this research was to increase the rate of hydrolysis and acidification in a two step reactor process by bioaugmentation with TWAS in an anaerobic or microaerophilic acidification stage. For this purpose initial acidification batch experiments (duplicate) were performed in 0.25/0.50 L serum bottles (0.15/0.40 L working volume) at 35 °C using a New Brunswick orbital shaker (100 rpm). Serum bottles contained 22.5% (v/v) acclimatized inoculum, diluted SKW with or without pretreatment and varying amounts of TWAS. Original inoculum was obtained from a mesophilic anaerobic digester at ROPEC municipal wastewater treatment plant (Gloucester, ON, Canada), which treats

combined primary and secondary sludge (5 d SRT). Inoculum was acclimatized to MW pretreatment over a period of 1.5 years in a mesophilic complete mixed digester, using SKW pretreated to 145 °C with a ramp of 2.7 °C/min operated at a HRT of 33–40 d. TWAS was supplemented in the serum bottles at 0, 15, and 30% based on VS (w/w) without any additional buffer and the food to microorganism ratio (F/M ratio) was in the same range as previously reported by Shahriari et al. (2012). For microaerophilic acidification, oxygen concentrations were maintained between 0.5 and 1 mg/L by bubbling air at 0.008 volume per volume per minute (vvm) at 35 °C through the serum bottles. All assays were monitored daily for VFA accumulation, pH and biogas (where applicable).

2.3. Semi-continuous reactors

A total of 12 mesophilic single and dual stage reactors (Fig. 1) were used to evaluate the effects of MW pretreatment, staging and HRT on the stabilization of the whole SKW waste (6 reactors) and free liquid extracted from whole SKW (6 reactors). The liquid fraction of SKW was evaluated to determine the feasibility of mesophilic AD of supernatant only. Free liquid was obtained by centrifuging whole SKW at 9725 RCF for 45 min followed by decantation of the liquid phase. Each group of 6 reactors was evaluated at HRTs of 20, 15, 12 and 9 d. Acidogenic reactors were operated at a fixed HRT of 2 d hence a 20 d HRT for a two stage system was 2 d/18 d for acidogenic (a) and methanogenic (m) phases respectively. HRT of 2 d for the first stage was based on the findings of the acidogenic fermentation assay described above.

Semi-continuous studies were performed using 0.5 L and 1.0 L Kimax bottles (0.3/0.6 L working volume) for acid and methane phase respectively and sealed with butyl rubber stoppers. Ports in the rubber stoppers with glass tubes were used to collect biogas and withdraw/add substrate once per day. Tedlar bags (Chromatographic Specialties Inc., ON, Canada) were used to collect biogas which was then released and measured using a manometer. A combination (50:50 V/V) of biomass from ROPEC (Shahriari et al., 2012) and granular biomass from Lake Utopia Paper (a chemical thermal pulp treatment plant located in St. George New Brunswick, Canada) was used as inoculum. This latter biomass was used in anaerobic baffled reactors to treat aircraft de-icing fluid previously (Kennedy and Barriault, 2005). The properties of the granular biomass are listed in Table 1. The reactors were kept on a shaker at 35 ± 1 °C and 90 rpm (PhycroTherm, New Brunswick Scientific Co. Inc., NB) and were started at an HRT of 20 d (2 d/18 d for the two-stage reactors) and were operated until they reached steady state (SS). Fluctuation of less than 10% in daily biogas production and completion of 2–3 HRTs was considered as SS. Reduction of HRT and concomitant increase in volumetric organic loadings were done slowly over a period of approximately 1 week in order to allow the microbial consortia to adapt to new conditions. Performance of semi-continuous reactors was assessed by monitoring pH, VFA, alkalinity, total solids (TS), VS, TCOD, SCOD and ammonia-N. Standard methods (APHA, 1995) and techniques given in Shahriari et al. (2011) were used for all analysis.

Table 1
Properties of granular biomass.

Parameter	Unit	Quantity
TCOD	mg/L	29.01 ± 0.38
SCOD	mg/L	0.84 ± 0.00
TS	(W/W)%	20.75 ± 0.09
VS	(W/W)%	16.01 ± 0.19
Alkalinity	mg $CaCO_3$/L	2108 ± 12
Ammonia	(NH_3-N) mg/L	212 ± 10
pH	–	8.27
TVFA	mg/L	50 ± 2

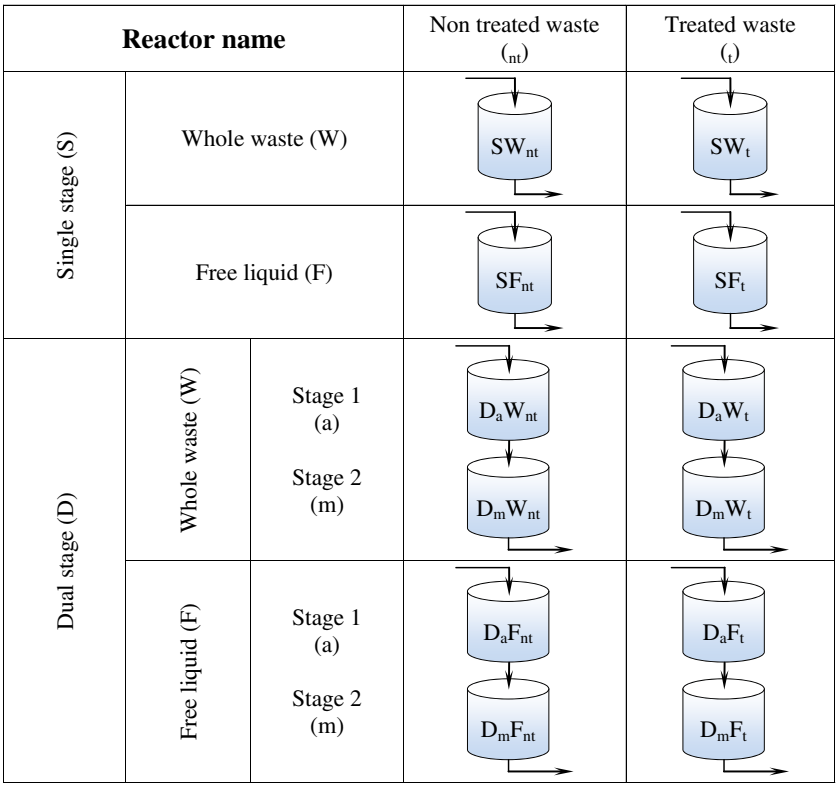


Fig. 1. Semi-continuous reactors.

3. Results

3.1. Acidogenic fermentation

Acidification was evaluated based on VFA accumulation (acetic, AA, propionic, PA and butyric acids, BA) over a period of 5 d. Average daily VFA accumulations as a measure of acidification are shown in Fig. 2a and b for anaerobic and microaerophilic conditions, respectively. For both anaerobic and microaerophilic acidification maximum VFA accumulation occurred after approximately 2 d and were 1700 and 1300 mg/L, respectively while the pH in the bottles decreased from 7.0 to about 6.0. No attempt to control/lower the pH was used in this evaluation of acidification. The pH values were slightly higher than desired for acidification especially with the assays supplemented with TWAS and this may be attributed to the higher alkalinity associated with the TWAS as well as consumption of the VFAs. Additionally, pH values in the microaerophilic reactors were likely higher (6.1–7.1) due to the bicarbonate buffering capacity caused by the formation of carbon dioxide during aerobic metabolism of substrate by facultative bacteria. Anaerobic acidification had both the highest VFA accumulation as well as the lowest pH values (6.0).

Jagadabhi et al. (2010) applied microaerophilic acidification for grass-silage at a microaeration rate of 0.0023 vvm and reported a 4 fold increase in VFAs production (2200–9300 mg/L) without any significant increase in cumulative SCOD in the effluent. VFA concentrations tended to decrease if microaeration was increased. Decreased VFA concentrations with increased microaerophilic were

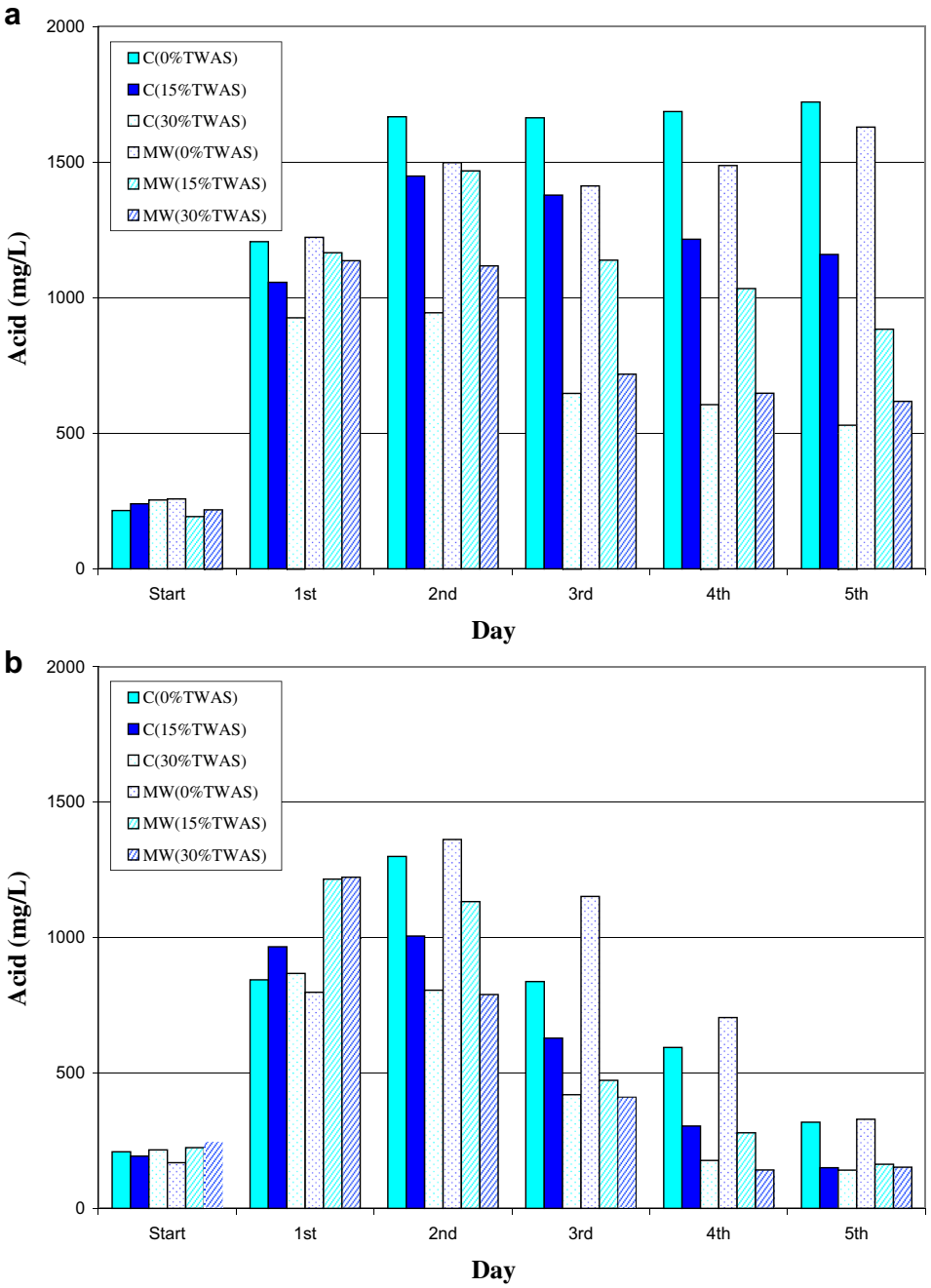


Fig. 2. (a) VFA accumulation under anaerobic condition. (b) VFA accumulation under microaerophilic conditions.

attributed to an increased rate of aerobic metabolism by facultative microbes. In the present study under microaerophilic conditions VFAs were consumed in days 3–5 and were less than 400 mg/L by day 5. In terms of TWAS addition the greater the percentage of TWAS the greater the consumption of VFAs. Additionally, microwaving of the SKW had negligible impact on acidification when compared to the controls. It would seem the combination of microaerophilic conditions and additional TWAS actually had a negative effect on acidification.

Anaerobic acidification in terms of acid accumulation was superior compared to microaerophilic acidification and produced VFAs concentration 6.8 times higher compared to initial controls. Unlike the microaerophilic conditions VFAs that accumulated were generally not consumed after day 2 and remained high till the end of the assay (day 5). It was noted that when TWAS was added there was some decrease in the concentration of accumulated VFAs from day 3–5 but not to the same extend as under microaerophilic conditions. When MW pretreatment was applied the VFA accumulation was high but slightly less (Day 2, 1499 mg/L) than without MW pretreatment and without TWAS addition. VFA accumulation coincided with maximum COD solubilisation (SCOD) which was also observed after two days for anaerobic acidification (without TWAS supplementation and MW pretreatment) which increased by 31% from 2.0 g/L to 2.6 g/L. It should be kept in mind that in this batch acidification assay both acidification and metabolism can occur simultaneously and that the increased accumulation of VFAs (and/or SCOD) is a strong indicator of the potential improvements that are possible in acidification. In comparison to a previous study (Shahriari et al., 2012) with the same conditions but with buffered media (pH = 7.0–8.0) cumulative biogas productions (CBPs) after 5 d in this study was less than 30% of the biogas produced in their study, indicating that some waste stabilization was still occurring during the assay, but at a slower rate. It had been hoped that pH adjustment could be avoided, as it has been reported that pH during acidogenesis can decrease towards a pH value in the range of 5–6.5 (Dinopoulou et al., 1988; Guerrero et al., 1999; Jagadabhi et al., 2010) depending on the nature of the substrate and organic load. Unfortunately, for the specific waste being evaluated this was not the case. This result indicates that in future acidification experiments lower pH conditions may be established to increase acidification and inhibit methanogenesis.

The fact that greater VFA accumulation occurred under anaerobic conditions is favourable for two stages AD, since the mixed liquor/effluent has a low oxidation reduction potential (ORP) and VFAs can be transferred directly to the methanogenic. Based on the evaluation of acidification of the specific waste to be treated (SKW with and without MW pretreatment) it was decided that for 2 stage digestion anaerobic acidification at an HRT of 2 d with initial pH adjustment to 5.0–6.0 would be used.

3.2. AD of microwaved SKW

Shahriari et al. (2011) have previously demonstrated that MW pretreatment increases solubilisation of SKW, which resulted in enhanced biogas production based on batch mesophilic biochemical methane potential (BMP) assays (Shahriari et al., 2012). This study evaluates the effects of pretreatment, reactor staging and HRT on stabilization and biogas production from whole SKW and free liquid extract from SKW. Over the duration of this study small changes in the characteristics of the SCOD/TCOD ratio of SKW (Table 2a and b) were observed. Small changes were likely related to differences in the nature of the raw materials used for sample preparation. SCOD/TCOD ratios were in the range of 0.38–0.44 and 0.42–0.51 for untreated and after MW pretreatment of SKW, respectively. Generally, SCOD of the SKW after MW pretreatment was approximately 15–26% and 12–37% greater than the controls for whole SKW and free liquid extract from SKW, respectively. For all test conditions prior to digestion the whole SKW and free liquid extract from SKW had ammonia-N (NH_3-N) below 40 mg/L; alkalinity of approximately 300 mg $CaCO_3$/L and VFAs of less than 300 mg/L.

Table 2a
Properties of feed at the different HRTs tested for whole waste.

SRT prop.	20 d		15 d		12 d		9 d	
	C	MW	C	MW	C	MW	C	MW
TS (%)	3.21	3.33	3.26	3.39	3.09	3.31	3.11	3.42
VS (%)	3.06	3.19	3.12	3.27	2.96	3.18	2.97	3.28
VS/TS	0.95	0.96	0.96	0.96	0.96	0.96	0.96	0.96
TCOD (g/L)	39.27	40.57	37.51	40.50	37.08	38.84	38.60	39.65
SCOD (g/L)	14.90	17.14	14.60	18.40	16.25	19.21	17.05	20.31
SCOD/TCOD	0.38	0.42	0.39	0.45	0.44	0.49	0.44	0.51

Semi-continuous tests were conducted in sequence from the longest HRT (20 d) to the minimum HRT of 9 d. This sequence was implemented to maximize acclimation of the anaerobic biomass and minimize risks of reactor instability due to high organic loads. It should be noted that the combined HRT of staged systems was the sum of the HRT of the methanogenic stage and the HRT of acidification reactor which was maintained constant at 2 d for the study. Initially, pH in the acidification stage reactors decreased to approximately 4.0. This was attributed to a combination of semi-continuous operation, high organic load and readily degradable substrate that resulted in accelerated acidification. In order to circumvent the low pH problem buffer was added daily to maintain a pH between 5.0 and 6.0 in the acidification stage which is the optimal pH range for acidogenic bacteria (Jash and Ghosh, 1996; Raynal et al., 1998). Once adjusted the VFAs in the acidification stage (2 d HRT) were in the range of 6000–14,000 mg/L based on the type of feed with an average AA/PA/BA distribution ratio of 25/10/65. Whole SKW had lower VFAs in the range of 6000–10,000 while for free liquid SKW concentrations were higher in the 7000–14,000 mg/L range. It was noted that VFA concentrations for MW pretreated samples of M-SKW were generally 1000–1500 mg/L greater than samples without MW pretreatment.

Solubilization of SKW can occur as a result of MW pretreatment or as a result of acidification in a single reactor or in the primary staging reactor. The question to be answered is, does pretreatment increase acidification or is it not required for SKW if staging is used. Comparison of effluent quality from primary stage acidification reactors indicated that maximum solubilisation occurred in reactor D_aW_{nt} which was fed untreated whole SKW. With a 2 d HRT the SCOD concentrations of the acid phase reactors with and without pretreatment were fairly similar suggesting that staging had the same impact as MW pretreatment in terms of solubilization. Over the course of the experiment (145 d) SCOD increases in the acid phase reactor treating non pretreated whole SKW ranged from 11 to 26% which was comparable to the 15–26% improvement as a

Table 2b
Properties of feed at the different HRTs tested for free liquid.

SRT prop.	20 d		15 d		12 d		9 d	
	C	MW	C	MW	C	MW	C	MW
TS (%)	2.38	3.32	2.52	3.24	2.43	3.08	2.23	3.04
VS (%)	2.20	3.13	2.34	3.06	2.23	2.89	2.01	2.87
VS/TS	0.93	0.94	0.93	0.94	0.92	0.94	0.90	0.94
TCOD (g/L)	27.5	37.0	28.07	36.08	30.86	33.30	26.23	34.80
SCOD (g/L)	23.0	31.50	25.82	32.36	28.74	32.25	21.35	28.20
SCOD/TCOD	0.84	0.85	0.92	0.90	0.93	0.97	0.81	0.81

result of microwaving of the whole SKW. This range of solubilizations is not unexpected based on the variation in characteristics of the SKW over the course of the experiment. It should be emphasised that the combination of MW pretreatment and primary acidification of the SKW resulted in no further improvement in SCOD although the VFA concentrations were slightly higher in the acid phase reactor with MW pretreatment. This initial finding suggests that MW pretreatment of SKW does not improve solubilisation when compared to no pretreatment and staging and only a marginal improvement in acidification based on VFA concentrations. It should be noted that this finding is related to SKW which is readily biodegradable and may or may not apply for real OFMSW which may have a larger component of less readily biodegradable material.

While MW pretreatment did not seem to have a significant positive impact on acidification it is possible that its benefits may be realized during methanogenesis in either the single or dual stage systems. It has been reported that MW pretreatment not only increases solubilisation of TWAS as a whole but the degree of pretreatment can influence the distribution of mass fractions of various sizes for both the soluble and suspended components. Eskicioglu et al. (2006) and Toreci et al. (2010), both noted an increase in the colloidal component of TWAS suspended solids after MW pretreatment and reported that the increased colloidal component improved the rate of biodegradation. Depending on the degree of MW pretreatment the nature of the soluble and suspended particle distributions can impact the overall AD performance, potentially leading to improved waste stabilization during methanogenesis which is best observed at high organic loading rates (OLRs) and short HRTs. Concomitantly, single and dual stage (phase separated) continuous mesophilic digesters were operated with and without MW pretreatment at increased loading rates at decreasing HRT of 20, 15, 12 and 9 d to assess the benefits of MW pretreatment and/or reactor configuration.

Tables 3–6 summarize the steady state characteristics of the digester effluents at different HRTs when fed with MW pretreated and untreated SKW. In order to evaluate the effects of pretreatment and reactor staging, results are compared to the single stage mesophilic digesters treating whole SKW without pretreatment (SW_{nt}) and the free liquid fraction of SKW without pretreatment (SF_{nt}). It should be noted that approximately 90% of full scale AD plants treating OFMSW in Europe are single stage digesters (De Baere, 2000). Results are compared and discussed in two sections (whole waste and free liquid fraction) to prevent confusion.

3.2.1. Biogas production of the whole waste

Daily biogas production data obtained from reactors fed with whole SKW with and without MW pretreatment throughout the experiment at the various HRTs are shown in Fig. 3a and the average of biogas production are given in Fig. 4a. MW pretreatment did not show a positive effect on methane production of the whole waste, when comparing single or two stage processes to each other with and without MW pretreatment. However, dual stage reactors performed significantly better in term of methane production and waste stabilization when compared to single stage digesters at similar HRTs both with and without MW pretreatment. The greatest relative increase in performance for the staged process without pretreatment (D_mW_{nt}) occurred at the shortest HRT of 9 d with methane production 60% greater compared to the single reactor control or single reactor with pretreatment as illustrated in Fig. 4b. The staged reactor combination achieved higher biogas production and organic removal efficiencies at all HRTs even at high OLR's when compared to the single stage systems. Additionally, the biogas yield (gas production per mass VS added) for D_mW_{nt} compared to the control (SWnt) was approximately 1.5 greater for all HRTs tested (Fig. 4c). Results are also in agreement with TCOD removal. Fig. 3c shows TCOD removal efficiencies for all conditions and clearly indicates the superior performance of the dual stage process without pretreatment (D_mW_{nt}) which tended to be about 10–35% greater than the single stage control system at similar HRTs. Similar improvements in biogas production and TCOD removal were not realized for SW_t and D_mW_t, especially at the shortest HRT and concomitant highest OLR. At an HRT of 9 d both single stage systems and the dual stage system with MW pretreatment had a TCOD removal efficiency of approximately 50% and 60% respectively, compared with about 70% for D_mW_{nt}. It should be noted that at HRTs of 20, 15 and 12 d D_mW_t always had TCOD removal efficiencies greater than the single stage reactor but were always slightly less than what was determined for D_mW_{nt} (Fig. 3c). Mesophilic AD staging without MW pretreatment of SKW was deemed to be the best and most economical configuration (D_mW_{nt}) to achieve the greatest biogas production, biogas yield as well as TCOD and VS removals over the OLRs and HRTs evaluated. Similar findings for the relative importance of AD staging vs. pretreatment has also been reported by Coelho et al. (2011) for TWAS digestion. It should be noted that for AD of TWAS improved TCOD removal and biogas production has been reported with high temperature MW pretreatment compared to controls in single stage digesters (Toreci et al., 2009), which is not the case in this study using SKW. However they had similar results when comparing dual stage reactors to each other and single stage digestion. It should be clarified; MW pretreatment of SKW did not improve the digestion of whole waste when comparing single or dual stage reactors to their counterpart especially at high OLRs. One reason can be the nature of the waste, SKW is highly biodegradable and there is little material that is difficult to hydrolyse and digest. Palmowski and Müller (2000) had similar findings to the present study when they investigated the effect of comminution of an organic waste on its anaerobic biodegradability, finding no improvement in biogas production after comminution pretreatment. In single stage systems operated at high OLR the OFMSW rapidly accumulates VFAs since methanogens grow slower than the acidogenic bacteria, which can cause an

Table 3
Steady state characterization of reactors at HRT of 20 d.

HRT = 20 d		Reactor							
Parameters	Units	SW_{nt}	SW_t	D_mW_{nt}	D_mW_t	SF_{nt}	SF_t	D_mF_{nt}	D_mF_t
OLR	g VS/L d	1.53	1.60	1.24	1.31	1.01	1.43	1.01	1.39
VS removal	%	74.7	78.5	87.3	83.5	70.9	79.3	80.6	80.9
Gas production	L/L d	1.20	1.22	1.02	1.06	0.78	1.18	0.80	1.07
CH_4 production	L/L g VS removed d	0.78	0.77	1.11	1.15	0.83	0.78	1.17	1.09
SCOD removal	%	92.1	93.6	97.1	95.3	97.5	98.2	95.2	93.1
TVFA	mg/L	649	126	497	485	68	46	882	1481
Alkalinity	mg $CaCO_3$/L	5025	4925	8050	8525	4650	4525	8625	10,000
pH	–	8.23	8.16	8.38	8.41	8.18	8.22	8.44	8.50

Table 4
Steady state characterization of reactors at HRT of 15 d.

HRT = 15 d		Reactor							
Parameters	Units	SW_{nt}	SW_t	D_mW_{nt}	D_mW_t	SF_{nt}	SF_t	D_mF_{nt}	D_mF_t
OLR	g VS/L d	2.08	2.18	1.66	1.90	1.47	2.09	1.33	2.06
VS removal	%	69.7	68.5	72.0	78.9	72.2	64.2	70.5	69.9
Gas production	L/L d	1.42	1.42	1.25	1.27	1.12	1.35	1.10	1.29
CH_4 production	L/L g VS removed d	0.73	0.72	1.48	1.17	0.80	0.72	1.52	1.13
SCOD removal	%	78.2	85.5	84.1	90.5	96.5	86.8	88.9	86.8
TVFA	mg/L	2204	1740	2286	1631	664	2888	1683	3243
Alkalinity	mg $CaCO_3$/L	7575	7381	8469	8638	7188	7450	6031	10,313
pH	–	8.12	8.06	8.21	8.30	8.02	8.02	8.33	8.34

imbalance in the anaerobic consortia and eventual reduction in methane production. Separation of the different AD populations and growth under more favourable conditions, has potential applications for OFMSW (Mohan and Bindhu, 2008).

Another possible reason for reduced waste stabilization after high temperature MW pretreatment is the formation of complex by products which may be refractory or inhibitory to methanogenesis. These compounds tend to have a greater negative impact at higher organic loadings and concomitant low HRTs and short contact times. Marin et al. (2010) pretreated SKW with MW at high temperature (175 °C) and reported decreased biogas production at heating rates of 3.9 and 1.9 °C/min. It was reported that decreased biogas production was likely related to complex compounds that were being produced at the high temperature. In another study (Shahriari et al., 2012), whole SKW pretreated at 115 °C and 145 °C showed 4–7% improvement in biogas production over untreated SKW (control) using batch mesophilic BMP assays. However, at 175 °C, biogas production decreased due to the formation of refractory compounds that inhibited digestion. In the present study, continuous AD treatment with MW temperature of 145 °C was implemented at progressively decreasing HRTs. At the long HRTs concentrations of refractory and/or inhibitory compounds in the reactor such as humic substances resulting from MW pretreatment are low and sufficient time is available such that little negative impact on digestion is observed. However, as HRTs are reduced (i.e. OLR increased) accumulation of humic and like compounds in the system increases while contact time for digestion decreases resulting in higher specific loading rates. Concomitantly the positive or negative impact of pretreatment on biogas production and waste stabilization becomes more evident as HRT is decreased and loading is increased. At decreasing HRTs there was a decrease in biogas production with MW pretreated waste compared to untreated waste. In this study higher concentrations of humic acid were observed after MW pretreatment, which were believed to have decreased biogas production compared to the untreated controls. At low HRTs and after microwaving there was sufficient substrate available but the microbial consortia were unable to metabolize the substrate rapidly which is indicative of non-competitive inhibition. Increasing the concentration of the substrate still does not allow the maximum enzyme activity rate to be reached which we speculate is happening in both the single and dual stage AD reactors.

3.2.2. Biogas production of the free liquid

Daily biogas production data obtained from reactors fed with the free liquid extracted from MW pretreated or untreated SKW throughout the experiment at the various HRTs are shown in Fig. 3b and the average of biogas production data are provided in Fig. 4a. Details pertaining to the concentration of the organics extracted into the free liquid per unit mass of whole SKW are found in Table 2b. It should be noted that SCOD concentrations of the free liquid extracted from MW pretreated whole SKW are 12–37% greater than free liquid obtained from untreated whole SKW. The volume of free liquid extracted per unit mass of MW pretreated whole SKW was also approximately 40% higher than the volume of free liquid obtained from controls. Since the free liquid extracted is based on per unit mass of whole SKW continuous reactors were operated at the various HRTs with the free liquid as produced. Fig. 3b indicates that per unit volume of free liquid treated, MW pretreatment results in higher biogas production compared to single and dual stage controls. Additionally, staging of reactors has a significant benefit in relation to the amount of biogas produced per unit volume of free liquid when compared to either of the single stage systems with or without MW pretreatment. The relative improvement for the 4 reactor configurations evaluated is best shown in Fig. 4b.

Maximum enhancement in relative CH_4 production when comparing SF_{nt} (354 ml/L d) to the other systems occurred at an HRT of 20 d and was 1.50, 1.46 and 1.97 fold greater in SF_t, D_mF_{nt} and D_mF_t respectively. While CH_4 production increased with decreasing HRT the relative change vs. the control reactor SF_{nt} tended to decrease. The staged process without pretreatment (D_mF_{nt}) produced biogas at the same rate or slightly higher as the single reactor with pretreatment (SF_t) at all HRTs (9–20 d). The fact that greater waste stabilization and CH_4 production occurred in D_mF_{nt} compared to SF_t is significant when considering that the organic

Table 5
Steady state characterization of reactors at HRT of 12 d.

HRT = 12 d		Reactor							
Parameters	Units	SW_{nt}	SW_t	D_mW_{nt}	D_mW_t	SF_{nt}	SF_t	D_mF_{nt}	D_mF_t
OLR	g VS/L d	2.46	2.65	2.29	2.34	1.95	2.55	2.19	2.80
VS removal	%	55.5	61.0	61.9	55.1	57.6	53.5	49.0	53.0
Gas production	L/L d	1.63	1.63	1.50	1.32	1.33	1.57	1.10	1.42
CH_4 production	L/L g VS removed d	0.90	0.75	1.40	1.34	0.90	0.87	1.32	1.24
SCOD removal	%	65.7	82.8	76.1	71.1	88.3	75.5	69.7	76.6
TVFA	mg/L	3453	2288	3151	3953	2443	4490	4872	6357
Alkalinity	mg $CaCO_3$/L	8417	8033	8467	8467	7850	8467	8933	10,150
pH	–	8.14	8.10	8.22	8.20	8.28	8.17	8.27	8.22

Table 6
Steady state characterization of reactors at tested HRT of 9 d.

HRT = 9 d		Reactor							
Parameters	Units	SW_{nt}	SW_t	D_mW_{nt}	D_mW_t	SF_{nt}	SF_t	D_mF_{nt}	D_mF_t
OLR	g VS/L d	3.32	3.67	3.42	3.26	2.49	3.23	2.66	4.09
VS removal	%	42.6	45.4	61.7	42.1	44.9	45.3	43.6	32.1
Gas production	L/L d	1.72	1.62	1.97	1.50	1.58	1.78	1.56	1.24
CH_4 production	L/L g VS removed d	0.92	0.73	1.31	1.50	1.03	0.92	1.86	1.31
SCOD removal	%	45.8	58.9	72.2	58.2	84.6	73.8	79.0	57.6
TVFA	mg/L	5720	5228	3366	5485	3332	5735	4177	9761
Alkalinity	mg $CaCO_3$/L	8950	8200	8900	8500	7950	8800	9100	10,400
pH	–	8.15	7.96	8.31	8.18	8.36	8.26	8.33	8.24

loading to SF_t was about 30–40% higher at each HRT evaluated (due to increased COD and VS concentrations in the free liquid extracted after MW pretreatment). This finding with regard to the benefit of staging vs. pretreatment alone is similar to what was reported above for the whole waste. Biogas results were also supported based on the COD removal data (Fig. 3d). When the HRT was decreased to 9 d (OLR increased) COD removal without pretreatment remained high (>60%), however for the single reactor with MW pretreatment COD efficiency decreased. It can also be noted that the COD removal efficiency also trended downward in D_mF_t tending to indicate that in this reactor as the HRT decreased there was some level of inhibition or increased recalcitrance of the free liquid as a result of MW pretreatment (discussed later). It should be reemphasised that the rates of biogas production were always greater than the control however the relative improvement decreased to 1.12 for SF_t and D_mF_t at an HRT of 9 d, compared to 1.50 and 1.97 at 20 d.

It may be difficult to clearly separate and understand the advantages of pretreatment and staging based on the volumetric parameters discussed above. However, Fig. 4c which is based on CH_4 production per unit mass of organics added clearly differentiates any advantages of either process. If one compares the yield of methane for single stage reactors with and without pretreatment over the 4 HRTs evaluated there is very little difference in the two systems. Similarly if one compares the dual stage systems with each other they are also fairly similar to each other except for the HRT of 9 d when D_mF_t began to fail and biogas production plummeted. The most important aspect of Fig. 4c supports the discussion above that when single reactors are compared to their dual stage counterparts at similar HRTs and loading rates there is a clear improvement in the yield of CH_4 production per unit mass of organics added for the dual stage system. On average the range of improvement over the HRTs evaluated when comparing like single and dual stage treatments were in the range of 25–75% (except D_mF_t). This again

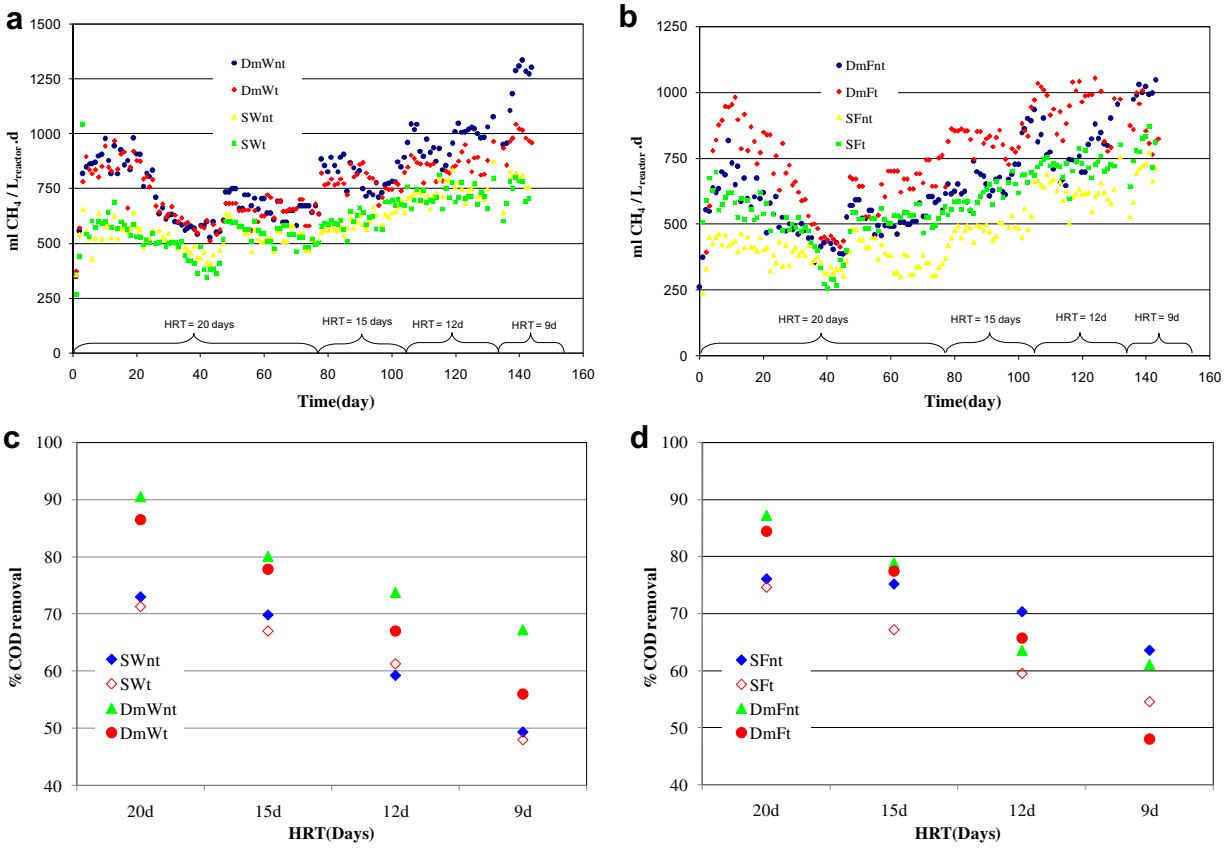


Fig. 3. (a) Daily CH_4 production for the whole waste. (b) Daily CH_4 production for free liquid. (c) Percentage of TCOD removal. (d) Percentage of TCOD removal.

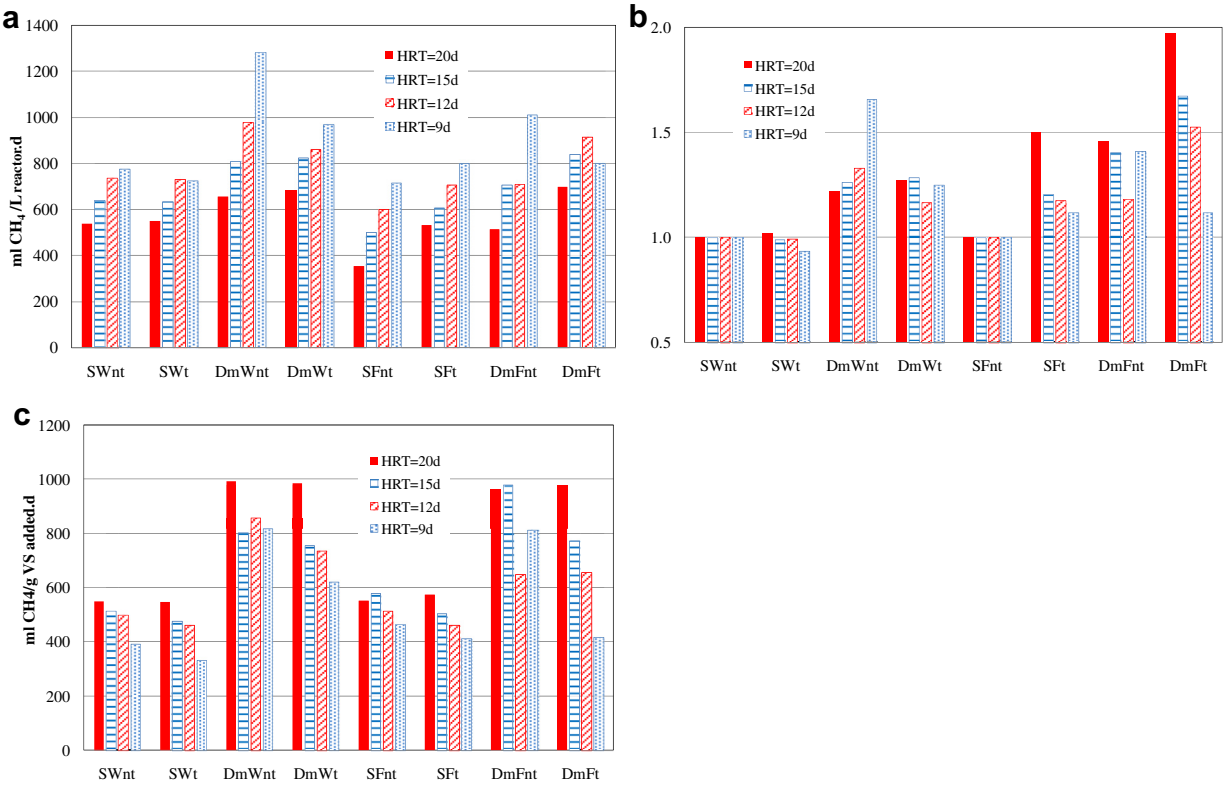


Fig. 4. (a) Average of daily CH_4 production. (b) Relative CH_4 production compared to controls. (c) CH_4 per g VS added per day.

indicates the advantages of staging on improving the digestion process for SKW. Additionally, the results also support the conclusion that staging provides more benefits than MW pretreatment. It should be remembered that MW pretreatment not only increased the concentration of organics in the free liquid but also the quantity of free liquid per mass of SKW treated. This will have an impact on the process if a mass balance approach is utilized.

Reduced methane production relative to SF_{nt} (control) at the shortest HRTs may be due to VFA accumulation possibly related to inhibition of methanogens. VFA of 1481 mg/L at an HRT of 20 d in D_mF_t increased to 9761 mg/L at an HRT of 9 d. Tables 3–6 show higher VFA accumulation in reactors with MW pretreated whole SKW and free liquid extracted from MW pretreated whole SKW compared to the controls in most cases. Approximately 65% of the VFAs produced in and transferred from stage one to the second stage was butyric acid. Among VFAs, propionic and butyric acid have been found to be the most inhibitory to methanogenesis (Mata-Alvarez, 2003; Warith et al., 2005). Callaghan et al. (2002) reported that with a VFA/alkalinity ratio of less than 0.4, AD is usually stable. For HRTs of 15 and 20 d this condition was met, but at an HRT of 12 d the ratios were 0.53, 0.47, 0.55 and 0.63 for reactors SF_t, D_mW_t, D_mF_{nt} and D_mF_t, respectively. At an HRT of 9 d D_mW_{nt} was the only process configuration with a ratio below 0.4, the rest were in the unfavourable range. The low VFA/alkalinity ratio is one likely reason for superior performance of D_mW_{nt} at short HRT (9 d). D_mF_t failed at an HRT of 9 d and had a VFA/alkalinity ratio of 0.94. SCOD results indicated that although increasing the OLR (shortening HRT) will provide a surplus of biodegradable substrate for methanogenesis, the microbial consortia in the two step reactor were not in equilibrium when treating the free liquid which resulted in VFA accumulation in the second reactor with concomitant reduced methane production.

The methane production has an inverse relationship with respect to the OLR (Fig. 4c). In general CH_4 produced/VS added decreased with shortening HRT. SCOD results showed that although increasing the OLR provides an abundance of organic substrate, the system is unable to use it as efficiently. At the 20 d HRT the maximum SCOD concentration (D_mF_t) was below 1500 mg/L, but it

Table 7
Observed and calculated values for methane production.

HRT reactor	20 d		15 d		12 d		9 d	
	Observed	Predicted	Observed	Predicted	Observed	Predicted	Observed	Predicted
SW_{nt}	0.32	0.29	0.38	0.38	0.44	0.38	0.47	0.45
SW_t	0.33	0.30	0.38	0.38	0.44	0.42	0.44	0.45
SF_{nt}	0.21	0.20	0.30	0.29	0.36	0.34	0.43	0.46
SF_t	0.32	0.27	0.36	0.35	0.42	0.38	0.48	0.43
D_mW_{nt}	0.35	0.32	0.42	0.39	0.49	0.44	0.60	0.54
D_mW_t	0.37	0.31	0.43	0.36	0.43	0.40	0.45	0.44
D_mF_{nt}	0.28	0.24	0.37	0.26	0.36	0.30	0.47	0.40
D_mF_t	0.38	0.29	0.44	0.38	0.46	0.39	0.37	0.37

Table 8
Methane production comparison (HRT = 20 d).

HRT = 20 d	Reactor							
	SW_{nt}	SW_t	D_mW_{nt}	D_mW_t	SF_{nt}	SF_t	D_mF_{nt}	D_mF_t
L CH4/L reactor per tonnes of waste	89.7	91.8	109.1	114.0	18.8	40.7	24.5	53.5
Improvement relative to SW_{nt}	–	2%	22%	27%	−21%	−45%	−27%	−59%
Improvement relative to SF_{nt}	477%	488%	580%	606%	–	116%	46%	185%

increased as HRT was shortened, and SCOD removal was 48% at an HRT of 9 d (majority of SCOD was VFAs). These results are in agreement with other similar studies which were reported by Mohan and Bindhu (2008) and Hartmann and Ahring (2006). Coelho et al. (2011) used MW pretreatment of TWAS to 96 °C and reported that gas production/mass VS added decreased with decreasing HRT with or without pretreatment. Toreci et al. (2009) also evaluated MW pretreatment of TWAS at temperatures above 100 °C and found that biogas production/mass VS added for controls decreased when HRTs were less than 10 d.

3.2.3. Mass balance

To evaluate the overall performance of the single and dual stage systems with and without MW pretreatment a COD mass balance was completed. COD conversion to methane was based on the theoretical conversion of 1 g COD removed per 350 ml methane produced (Speece, 1996). The daily conversion of COD is known (ΔCOD), so CH_4 production can be calculated and compared with the observed values (Table 7). While most of the calculated results are in agreement with the experimental observations it could be argued that the observed values were slightly higher than predicted. However based on daily fluctuations in feed concentration, experimental determination of COD values and errors in measurement of gas the data comparison shows a very close correlation and gives us added confidence in our evaluation of the various processes and MW pretreatment and the conclusions we have drawn.

Under what was deemed to be the best condition (HRT = 20 d) for AD of SKW, volumetric methane production yields were determined for the various pretreatment and reactor configurations and are shown in Table 8. The best condition was selected based on the highest methane yield per mass of SKW volatile solids added as well as reactor stability and taking into account provincial regulations for treatment of municipal sludge which are set at 15 d.

Biogas production values were determined per volume of SKW treated (approximately 16% solids) and also considering the mass and concentration of free liquid extracted from whole SKW. For semi continuous AD the most economical combination to enhance biogas production from SKW is digestion of whole waste SKW in a two stage configuration without pretreatment which had a 22% improvement vs. the control. The expenditure of energy for pretreatment would not justify the application of D_mW_t which had a 27% improvement vs. the control but only 4% improvement vs. D_mW_{nt}. Treatment of the free liquid only is not a practical option as in the best case scenario (D_mF_t) it only accounts for 59% of the methane production of the whole waste compared to control (SW_{nt}). Concomitantly, pretreatment of SKW does not release sufficient organics in to the free liquid phase to justify treatment of the liquid phase only. However, if digestion of the liquid fraction is desired, pretreatment at 145 °C and staging significantly increases biogas production from this fraction. However, methane produced from the free liquid in SF_t (single reactor after MW pretreatment) and D_mF_t (dual state after MW pretreatment) were approximately two and three fold greater than the control reactor (SF_{nt}). Additionally, both single and dual stage systems treating free liquid produced more methane than the dual system without pretreatment (D_mF_{nt}) which can be attributed to the higher concentration of organics and volume of free liquid released from the whole SKW after MW pretreatment. If one just looks at the impact of staging vs. single reactors (SW_{nt} vs D_mW_{nt} or SF_{nt} vs D_mF_{nt}) a 22% and 46% improvement respectively can be obtained by selection of an alternate 2 stage reactor configuration.

4. Conclusion

Maximum acidification of SKW under anaerobic condition was determined to be 2 d based on a modified BMP assay which was selected for the HRT of the acidification stage of the dual digestion system and was found to be appropriate.

For digestion of the whole SKW or the free liquid obtained from SKW, staging of the AD process had a greater positive impact on waste stabilization and improved methane yield compared to single stage reactors or MW pretreatment. For SKW which was deemed to be composed of readily biodegradable components it is recommended that staging be utilized without MW pretreatment as being the most economical mode for AD. This conclusion may be challenged if the OFMSW has a greater proportion of more difficult or less biodegradable components which would justify MW pretreatment. However, in both cases (SKW or OFMSW) dual stage AD should be considered.

MW pretreatment increased the solubilization and volume of free liquid from SKW. However, pretreatment and digestion of the free liquid fraction only, is not a sustainable option.

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